

PlanSpec

An open standard for structured AI execution plans.

The Problem

When an AI agent plans work — writing code, modifying infrastructure, executing a multi-step task — that plan exists as a chat message. Unstructured. Unverifiable. Impossible to audit after the fact.

As AI agents take on higher-stakes work in enterprise environments, this becomes a critical gap:

- **No accountability** — there's no record of what the AI intended to do, what constraints it acknowledged, or what risks it identified
- **No validation** — there's no way to check if the plan is complete, consistent, or safe before execution begins
- **No oversight** — there's no boundary between what the AI is authorized to do autonomously and what requires human approval
- **No audit trail** — when something goes wrong, there's no artifact to trace the decision chain

The Solution

PlanSpec defines a standard format for execution plans that are:

- **Human-readable** — a senior engineer or manager can review one without AI assistance
- **Machine-executable** — structured enough for an agent to follow step by step
- **Validatable** — constraints, success criteria, and scope boundaries are explicit and checkable
- **Auditable** — every decision has recorded rationale, every revision has history

What's in a PlanSpec

A PlanSpec document captures six critical dimensions of any execution plan:

Dimension	What It Answers
Intent	What are we doing and why? What does success look like?
Scope	What is explicitly in and out of bounds? What assumptions are we making?
Approach	What are the steps, their dependencies, and their complexity?
Constraints	What conditions must hold throughout execution? (Security, compliance, performance)
Delegation	What can the AI do autonomously vs. what needs human sign-off?
Risks	What could go wrong, how likely is it, and what's our mitigation?

Every PlanSpec is versioned, revision-tracked, and carries a full decision history.

Why an Open Standard?

Today, every AI planning tool invents its own format. Plans can't be shared between tools, reviewed by standardized processes, or audited against a common baseline.

An open standard means:

- **Tool interoperability** — any planning tool can produce PlanSpec documents; any execution tool can consume them
- **Independent validation** — a PlanSpec can be reviewed by a human, a tool, or a committee using consistent criteria
- **Organizational governance** — enterprises can adopt PlanSpec as a policy: "no AI agent executes without an approved PlanSpec"
- **Ecosystem growth** — validators, dashboards, audit tools, and comparison engines all build on the same format

Real-World Example

Robotics in Industrial Settings

A robotics company deploying autonomous systems in a refinery needs to ensure every AI-planned operation is reviewed before execution. A PlanSpec document specifies the scope (valves, sensors, control systems), safety constraints (tolerances, regulatory compliance, rollback procedures), delegation boundaries (autonomous vs. human-approved steps), measurable success criteria, and risk mitigations. The plan is reviewed by a safety engineer, approved, and handed to the execution system. If anything goes wrong, the PlanSpec is the artifact investigators use to trace decisions.

Who Should Care?

Audience	Why
Engineering leaders	Standardize how AI agents plan and execute work across teams
Security & compliance	Gain auditability and oversight over AI-driven operations
AI tool builders	Adopt a proven format instead of inventing your own
Enterprise architects	Define governance policies for AI agent deployment
Regulators & auditors	Establish a baseline artifact for AI accountability

Current Status

- **Version 1.2.0** published and available under MIT license
- Full specification, JSON Schema, and examples at github.com/Dazlarus-Industries/planspec
- Reference implementation (Planclave engine) in development by Dazlarus Industries
- Seeking early adopters and feedback from organizations deploying AI agents in production